COMSATS University Islamabad, Lahore Campus

☑ Midterm Examination – FALL 2023

Course Title:	Design and Analysis of Algorithms			Course Code:	CSC301	Credit Hours:	3(3,0)
Course Instructor:	Dr. Aasim Qureshi, Dr. Hasan Jamal			Programme Name:	BS Computer Science		
Semester:	5th	Batch:	FA21-BCS	Section: A, B, C	Date:	04/11/2023	
Time Allowed:	**90 Minutes**			**Maximum Marks:**		**50**	
Student's Name:	[redacted]			Reg. No.	[redacted]		

Important Instructions / Guidelines:

- Be precise and to the point while answering any question.
- Show all your work, as partial credits will be given. You will be graded not only on the correctness of your answer, but also on the clarity with which you express it. Please be neat.
- **Good luck!**

Question 1: **CLO: <1>; Bloom Taxonomy Level: <Understanding>** **[Marks: 5]**

Why theoretical analysis of algorithms is preferred over empirical analysis?

Question 2: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 10]**

Prove or disprove: $n^3 + 5n^2 - 2n + 8 \in \Theta(n^3)$

Question 3: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 10]**

Provide a line-by-line analysis and construct function T(n) that give the runtime of MY-FUNCTION() as a function of "n". Also determine the Big-Oh of MY-FUNCTION().

```
MY-FUNCTION()
    for i = 0 to n
        sum = foobar(i)

BAR(a)
    for i = 0 to n*n
        sum += i * a
    return sum

FOO(a)
    return a * bar(a)

FOOBAR(a)
    for i = 0 to n
        sum = foo(a) * i
        return sum
```

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Question 4: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 2.5 x 4 = 10]**

In not more than five lines each, provide the answers to the following questions. Be reasonable in your assumptions. Your answer must appear in the first few words, followed by strong justification to support your answer.

a) You are given an unsorted list of 10,000 integers with repeating numbers ranging from 5000 to 9,000. Which sorting algorithm will you choose to sort this data set? Justify your answer.

b) Suppose you have found a piece of code from "Stackoverflow" that takes a list of sorted numbers as input and perform some operations on the sorted list. However, unfortunately, there is a bug in the code that randomly shuffles a few numbers. To save time, instead of fixing the bug, you decide to sort this list again. Which algorithm will you choose to sort this list? Justify your answer.

c) You have a uniformly distributed set of unsorted integers of 7,000 numbers, within the range 0 to 10,000. Which algorithm will you choose to sort this data set? Justify your answer.

d) You have a list of share prices of 5000 stocks. To sort this list, you need a stable algorithm with an absolute guarantee of O(n lg n) performance in the worst case. Which algorithm will you choose? Justify your answer.

Question 5: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 10]**

Solve the following recurrence using the "Recursion Tree Method".

$$T(n) = 2T\left(\frac{n}{4}\right) + n^2$$

Question 6: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 5]**

Solve the following recurrence using the "Master Theorem Method".

$$T(n) = 6T\left(\frac{n}{2}\right) + n^3$$

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